2	**a**	Use of *mgh = ½ m* v^2 ***and*** *makes h subject* h = 14.7 or 15 m	 A1
	bi	Calculate the final vertical velocity at C (using v = 0 + at=9.81x1.6) v = 15.7 or 16 m s^{-1}	C1 A1
	bii	Straight line of positive gradient) Starting at 0 ms^{-1} and ends at 16 ms^{-1} at 1.6 s.	 A1
	biii	Use of pythagoras' theorem: resultant $v^2 = 15.7^2 + 17^2$ v = 23 or 23.1 m s^{-1}	M1
	c	slope: smaller change in vertical component of velocity/ smaller change in vertical component of momentum by Newton's second law, the force experienced = rate of change of momentum is less, so less risk of injury	M1 A1
3	**ai**	Archimedes' Principle states that the upthrust on a body completely or	B1